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Species composition of phlebotomine sand fly fauna in an area with sporadic cases of *Leishmania infantum* human visceral leishmaniasis, Morocco

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ABSTRACT

Visceral and cutaneous leishmaniases are the main endemic vector-born diseases in Morocco. Human visceral leishmaniasis (HVL), by *Leishmania infantum*, currently presents a significant health problem throughout the country and may constitute factor for death, especially among children with less than 15 years old. In the past, HVL has been basically absent or at least sporadic in Marrakesh-Tensift-Al Haouz region; however it became significant during the last decade. An entomological survey and a retrospective study on *L. infantum* HVL cases had been carried out to assess the risk of the disease apparition in this region.

7046 sand flies (Diptera: Psychodidae) were collected and studied from twelve localities within Marrakesh-Tensift-Al Haouz region. The result shows the presence of ten sand fly species, 58.76% from the genus *Phlebotomus* and 41.24% from genus *Sergentomyia.* A further analysis indicates that *Phlebotomus perniciosus, Phlebotomus longicuspis* and *Phlebotomus ariasi* species, incriminated vectors of *L. infantum*, are dominant (35.56%), so, we describe their spatial (according to altitude and biotopes) and temporal (seasonal activity) distribution in study area.

1. Introduction

In the Mediterranean countries, *Leishmania infantum* (Kinetoplastida: Trypanosomatidae) is the etiologic agent of Human Visceral Leishmaniasis (HVL). Most of its foci in the Old World have a Mediterranean climate and sand fly vectors, usually *Phlebotomus* (*Larroussius*) species, can diapauses. Ready (2014) suggests that there is a risk of the disease spreading with climate change.

In Morocco, *L. infantum* is widespread in all regions; however it is more frequent in the northern part with sporadic cases observed in the south (Rhajaoui, 2009). It can also cause Human Cutaneous Leishmaniasis (HCL) with the dog as the common reservoir host for both forms (Kahime et al., 2014). *L. infantum* is transmitted by *Phlebotomus ariasi* and *Phlebotomus perniciosus* (Diptera: Psychodidae), its incriminated vectors in the Mediterranean area (Benabdennbi et al., 1999; Amro et al., 2013), and potentially by *Phlebotomus longicuspis* (Pesson et al., 2004).

In northern Morocco, the first case of HCL due to *L. infantum* was reported in 1996, in the Taounate province, within an active focus of *L. infantum* HVL (Rioux et al., 1996). Recently, a new focus of *L. infantum* HCL was described in the Sidi Kacem province (Rhajaoui et al., 2007).

Regarding HVL, its evolution is progressing in Morocco with 154 cases declared in 2004 against 76 cases in 1997 (Tamimy, 2011). The incidence of this form was approximately 150 cases/year between 2006 and 2008 (Alvar et al., 2012) of which children under 4 years old are most affected (Amro et al., 2013). Almost all of these cases originated in traditional northern foci such as Chefchaoun, Taounate, Taza, Fes, My Yacoub, Meknes, Sefrou, Al Houceima and Sidi Kacem (Fig. 1).

In central and southern Morocco, including the Marrakesh-Tensift-Al Haouz region, *L. infantum* infections are sporadic. However, their appearance was recently noted to be more frequent (imported and/or autochthonous cases) (Moroccan Ministry of Health, 2014). In addition, Dereure et al. (1986) demonstrated the presence of an autochthonous canine visceral leishmaniasis cycle (by *L. infantum*) in the High and Anti-Atlas mountains. Recently,

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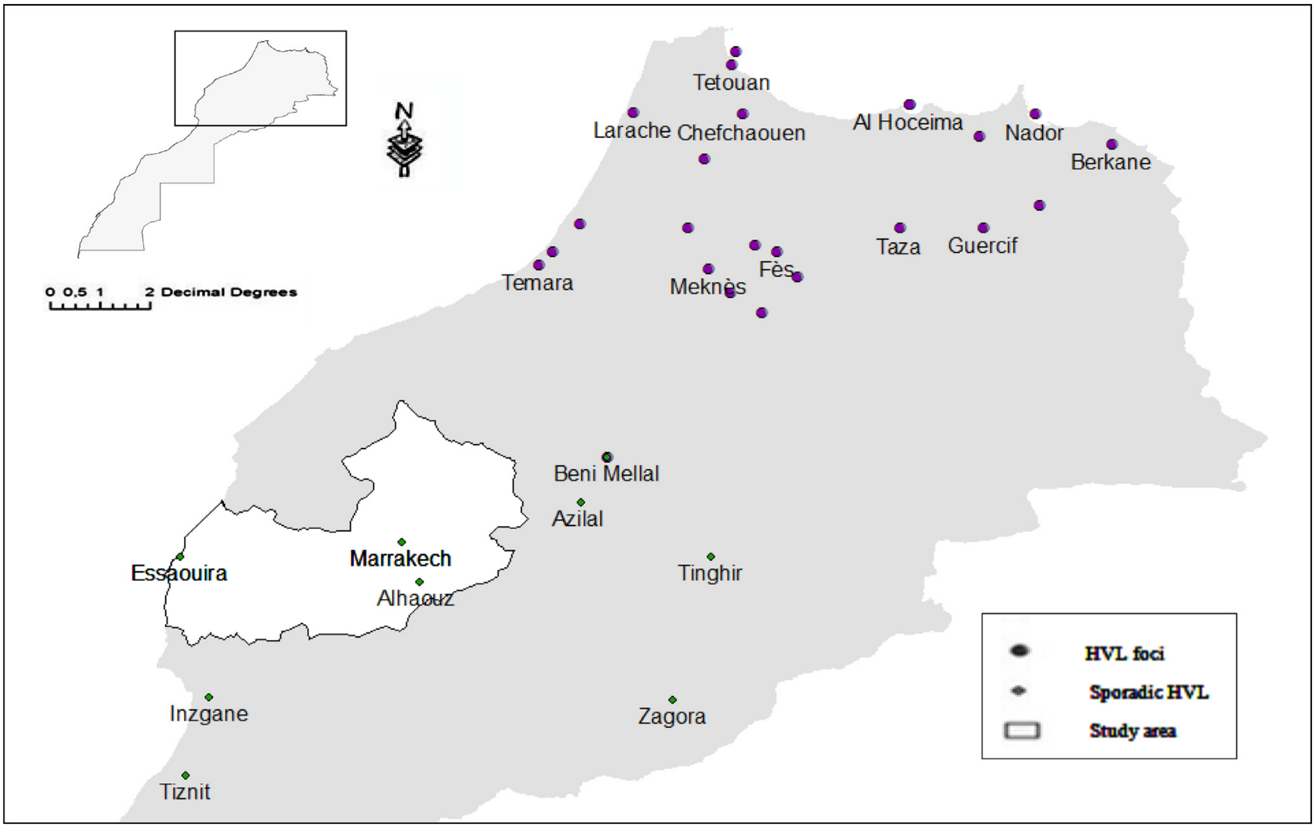

Fig. 1. Geographical distribution of human visceral leishmaniasis (HVL) due to *L. infantum* in Morocco (Moroccan Ministry of Health, 2011) with study area localization.

Boussaa et al. (2014) highlighted the presence of canine leishmaniasis infection in the southern regions, including our study area, with high seroprevalence, respectively, 81.8% and 87.8% by Elisa and Western blot tests.

As shown in Fig. 1, two areas in 2011 were exposed to *L. infantum* HVL in Morocco. The first one exists mainly along a north-east axis where *L. infantum* is endemic, whereas the second area in the center and the south, includes the Marrakesh-Tensift-Al Haouz, Tadla-Azilal and Agadir-Tiznite regions, as well as the southeastern area (including Zagora, Tinghhir and the entire area below the High Atlas Mountains) where many *L. infantum* HVL cases were recorded.

The study of ecology and epidemiology of leishmaniasis are important measures for the management and planning of disease control. The entomological survey accompanied by epidemiological data is a major component for combating against disease. In our study area, a number of *L. infantum* HVL cases have been recorded during the last decade, especially in Al Haouz Province (Table 1). Therefore, our aim is to assess the risk of HVL in this area through describing its sand fly fauna and studying the seasonality of its potential vectors.

2. Materials and methods

2.1. Study area

Field studies were conducted in the plain of Al Haouz of Marrakesh and in the High-Atlas Mountains (Fig. 1). Twelve stations were prospected with an altitude ranging from 500 to 1800 m above sea level (Table 2). In these areas, the climate is arid to semi-arid. The monthly mean temperatures vary between 37.7 °C in July and 4.9 °C in January, and the mean annual rainfall is low (less than 400 mm per year in the plain and about 600 mm per year in the High-Atlas Mountains).

In Al Haouz plain, the vegetation, dominated by steppe formations, consists essentially of *Zizyphus lotus* (jujube), *Retama monosperma* (retam), *Pistacia atlantica* (pistachio), *Acacia gummifera* (gum) and Chenopodiaceae *Salicornia arabica*, *Atriplex* sp., *Haloxylon scoparium*). The slopes of the High Atlas are forested by various species (juniper, oak, cedar). Lower valleys are characterized by the olive (*Olea europaea*), Oleaster, Arbor-vitae, the *Juniper phoenicia* and Mastic. In very sheltered valleys live the oxycedre and Aleppo pine and the Xeric thorny is the only persistent vegetation in the high mountains. The total population is 3102,652 inhabitants, composed mainly of farmers (Table 1), and most households suffer from poor sanitation.

2.2. Epidemiology

It is a retrospective analysis of the HVL by *L. infantum* situation in Marrakesh-Tensift-Al Haouz region. Epidemiological data were obtained from the bulletins, registers and reports published by the local and national medical services. These epidemiological data were recorded after active (especially for CL) or passive (leishmaniasis is a certifiable disease in Morocco) screening. Suspected cases with typical symptoms (fever, losing weight, splenomegaly, hepatomegaly . . .) were confirmed by parasitological and serological (IFI or Elisa) analysis (Moroccan Ministry of Health, 2010).

2.3. Entomology

Specimens were captured overnight with castor oil impregnated A4 papers (sticky traps). In twelve stations, 30/night traps were

Table 1
Demographic (High Planning Commission of Morocco, 2014) and Epidemiological (*L. infantum* HVL cases recoded by Moroccan Ministry of Health, 2014) data of Marrakesh-Tensift-Al Haouz region.

Province	Urbanization (%)	Population	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	Total/site
Al Haouz	10.78	484,312	2	3	2	3	1		1		1	1	14
Chichaoua	12.91	339,818	1	1	1		1		1	2			7
Marrakech	78.78	1070,838				1	1		1	1	1		5
Essaouira	21.10	452,979					1		1		2		4
Kelâa des Sraghnas	0.004	754,705											0
		Total/year	**3**	**4**	**3**	**4**	**5**	**0**	**5**	**3**	**2**	**1**	**32**

placed simultaneously in various biotopes (Table 2) and removed early in the morning (transect of twelve stations).

For *Larroussius* subgenus seasonality, Ourika, Zaouiate Bouhouta and Yabora, stations seemed to be more interesting in terms of abundance and frequency. So, in addition, for each station, 30 sticky traps were placed, on one night by month, in animal shelters (cow pen) during one year study (January to December 2012).

In laboratory, caught specimens were stored at 70 °C ethanol. They were cleared, dehydrated, colored with fuschin 3‰ and mounted in Canada balsam after dissection.

Morphological identification was made according to the shapes of the pharynges, cibarial teeth, female spermathecae and male genitalia, with the help of the keys and descriptions published by the Moroccan Ministry of Health (1997). For species of subgenus *Larroussius*, the morphological identification of males is based on their genital apparatus (Benabdennbi and Pesson, 1998; Benabdennbi et al., 1999; Guernaoui et al., 2005a), while females are separated according to the form and position of the dilatation on the spermathecal duct (Léger et al., 1983; Killick Kendrick et al., 1991).

2.4. Data analysis

In order to illustrate the sand fly populations in different sites, we calculated: (i) density of sand flies collected – number of specimens/m^2 of sticky traps/night; (ii) relative frequency – number of specimens of species (x_1)/total number of specimen × 100; and (iii) species richness – number of species in each station.

Data entry and analysis were conducted using the Statistical Package for the Social Sciences (SPSS, Version 10.0). The differences between studied variables were determined through comparison of means established by the one-factor analysis of variance (ANOVA) and the general linear model for multivariate analysis. Thus,

Table 2
Ecological and entomological data in the 11 study sites..

Locality	Altitude (m)	Urbanization types	Dominant vegetation	Species richness	Biotopes	Dominant sand fly species
Oulad Yahia	572	Peri-urban	*Acacia gunifera*	**5**	Peridomestic	*Phlebotomus longicuspis*
Amkhlij	846	Rural	*Euphorbia resinifera*	**9**	Domestic Peridomestic Barbacane	*P. papatasi* *P. sergenti* *Sergentomyia fallax*
Ourika	850	Rural	*Olea europea; Olea europaea var. oleaster*	**9**	Domestic Peridomestic Cow pen Barbacane	*P. longicuspis* *P. sergenti* *P. perniciosus* *S. fallax*
Aagreb	873	Rural	*Olea europea; Olea europaea var. oleaster*	**8**	Domestic Peridomestic Sylvatic	*P. sergenti* *S. minuta* *S. fallax*
Oulmès	900	Rural	*Juglans regia*	**2**	Sylvatic	*S. minuta*
Zaouit Bouhouta	969	Peri-urban	*Eucalyptus camaldulensis*; *Olea europea*	**8**	Domestic Peridomestic Sylvatic Cow pen	*P. papatasi* *P. longicuspis* *P. ariasi* *P. perniciosus*
Tnin Khmiss	895	Rural	*Olea europea; Olea europaea var. oleaster; Euphorbia resinifera Ceratonia siliqua*	**8**	Domestic Peridomestic	*P. longicuspis* *S. fallax*
Akhlij	917	Rural	*Olea europea*; *Olea europaea var. oleaster*	**9**	Domestic Peridomestic	*P. perniciosus* *S. fallax*
Tahanaoute	1000	Urban	*Eucalyptus camaldulensis.*	**8**	Domestic Peridomestic Sylvatic	*P. sergenti* *P. perniciosus* *S. fallax*
Yabora	1155	Rural	*Olea europea*; *Ceratonia siliqua*	**8**	Domestic Peridomestic Cow pen Sylvatic	*P. longicuspis* *P. ariasi* *P. perniciosus* *S. minuta*
Setti Fadma	1210	Rural	*Juglans regia*; *Juniperus oxycedrus*	**7**	Peridomestic Sylvatic	*P. sergenti* *S. fallax*
Imlil	1637	Rural	*Juglans regia*; *Olea europea L.*	**3**	Garbage	*P. perniciosus*

Table 3
Numbers of males (M), females (F) sand flies collected and the (%) of each species by site.

		Oulad Yahia	Amkhlij	Ourika	Aagreb	Oulmès	Zaouit Bouhouta	Tnin Khmiss	Akhlij	Tahanaoute	Yabora	Setti Fadma	Imlil
Phlebotomus perniciosus	**M**	10	55	201	9	1	80	43	94	8	835	19	44
	F	0	5	23	7	1	78	4	7	3	29	2	0
	%	13.16	9.09	9.95	3.17	28.57	32.44	6.62	26.72	9.65	52.08	18.26	52.38
P. longicuspis	**M**	15	44	150	1	0	78	74	31	3	120	13	15
	F	12	9	17	0	0	61	6	2	0	8	0	0
	%	35.53	8.03	7.42	0.20	–	28.54	11.27	8.73	2.63	7.72	11.30	17.86
P. ariasi	**M**	0	4	7	0	0	46	3	0	2	178	0	25
	F	0	3	4	0	0	0	0	1	0	10	6	0
	%	–	1.06	0.49	–	–	9.45	0.42	0.26	1.75	11.33	5.22	29.76
P. sergenti	**M**	2	117	320	105	0	20	65	33	49	47	20	0
	F	1	29	58	19	0	35	9	1	9	5	11	0
	%	3.95	22.12	16.79	24.60	–	11.29	10.42	8.99	50.88	3.13	26.96	–
P. kazeruni	**M**	0	0	0	0	0	4	0	0	0	2	0	0
	F	0	0	0	0	0	0	0	0	0	0	0	0
	%	–	–	–	–	–	0.82	–	–	–	0.12	–	–
P. alexandri	**M**	0	4	11	3	0	0	0	4	4	0	0	0
	F	0	0	0	0	0	0	0	0	0	0	0	0
	%	–	0.61	0.49	0.60	–	–	–	1.06	3.51	–	–	–
P. papatasi	**M**	14	140	245	35	0	46	27	43	2	12	5	0
	F	6	14	25	3	0	17	3	5	3	2	0	0
	%	26.32	23.33	11.99	7.54	–	12.94	4.23	12.70	4.39	0.84	4.35	–
Sergentomyia minuta	**M**	0	18	168	39	0	6	85	26	8	197	12	0
	F	0	22	170	30	5	3	88	30	4	112	21	0
	%	–	6.06	15.01	13.69	71.43	1. 85	24.37	14.81	10.53	18.63	28.70	–
S. fallax	**M**	14	73	332	119	0	12	101	39	11	33	6	0
	F	2	120	515	133	0	1	201	61	8	69	0	0
	%	21.05	29.24	37.61	50.00	–	2.67	42.54	26.46	16.67	6.15	5.22	–
S. dreyfussi	**M**	0	1	3	1	0	0	0	1	0	0	0	0
	F	0	2	3	0	0	0	1	0	0	0	0	0
	%	–	0.45	0.27	0.20	–	–	0.14	0.26	–	–	–	–
	Total	**76**	**660**	**2252**	**504**	**7**	**487**	**710**	**378**	**114**	**1659**	**115**	**84**

multiple *a posteriori* comparison tests were performed to determine homogeneous groups using the Tukey's test (with 5% significance level).

3. Results

In this entomological survey, a total of 7046 specimens belonging to ten species were collected, seven of *Phlebotomus* genus (representing 58.76% of the total catch) and three of *Sergentomyia* genus (41.24%). Table 3 summarizes the number of males (M), females (F), and the percentage (%) of each species.

Among the *Phlebotomus* species caught, *P. perniciosus* was the most predominant (37.63%) species, followed by *Phlebotomus sergenti* (23.07%), *P. longicuspis* (15.92%), *Phlebotomus papatasi* (15.63%), *P. ariasi* (6.98%), *Phlebotomus alexandri* (0.63%), and *Phlebotomus kazeruni* (0.14%). However, for *Sergentomyia* genus, the *Sergentomyia fallax* represents (63.66%), followed by *Sergentomyia minuta* (35.93%) and then the *Sergentomyia dreyfussi* (0.41%). *P. kazeruni* was collected for the first time in this area, exclusively at Zaouiate Bouhouta and Yabora.

Our observation shows no obvious morphological anomalies among collected sand flies. In contrast, all the male *P. perniciosus* were considered to be of a slightly atypical form (PNA). This atypical morph of male *P. perniciosus* showed single tipped copulatory valves curved at their apex with a number of coxite hairs up to 14.

For each *Phlebotomus* species the ratio of male to female (males/females >4), and for *S. minuta* (males/females >1), the males caught outnumbered the females. However, for *S. fallax* the ratio of male/female is 0.67, whereas *S. dreyfussi* had an equal sex distribution.

The sand fly fauna of Ourika, Amkhlij and Akhlij showed the greatest biodiversity (9 out of 10 species were collected) while that of the Oulmes showed the least (2/10 species are collected) (Table 2). In terms of sand fly individuals, Ourika (2252 specimens) and Yabora (1659 specimens) are the most important stations compared to Oulmes, where only seven individuals were collected for one night (Table 3).

The prevalence of *Larroussius* species was compared in domestic, cow pen and sylvatic biotopes in 572–900 m altitude ranges and 917–1637 m altitude ranges (Fig. 2). According to biotopes, *Larroussius* species individuals were lower in sylvatic habitat. In contrast, and in term of altitude, *Larroussius* species individuals were very important in 917–1637 m altitude. *Larroussius* species distribution showed a low positive correlation with the altitude, but we noted that *P. ariasi* is more sensitive to altitude ranges ($r = 0.3$), followed by *P. perniciosus* ($r = 0.2$) and *P. longicuspis* ($r = 0.1$).

Fig. 3 represents the seasonal trends of *P. perniciosus*, *P. ariasi* and *P. longicuspis* in Ourika (850 m), Zaouiate Bouhouta (969 m) and Yabora (1155 m) during one year study. According to ANOVA analysis, we noted significant differences between the three stations ($p < 0.05$). Yabora station is suitable for the development of *Larroussius* species (mainly *P. perniciosus* and *P. ariasi*), while, Ourika station showed the low density of these three *Larroussius* species.

4. Discussion

For an epidemiological and ecological point of view, Marrakesh-Tensift-Al Haouz region offers an interesting field to study. Its position in a junction zone between *L. infantum* foci in the north region (Rhajaoui, 2009; Kahime et al., 2014) and those of *L. major* in Ouarzazate and Errachidia provinces (Rhajaoui et al., 2007). Its climate is characterized by several stages, from Mediterranean in the west to arid and semi-arid. Similarly its geographical position includes some plains in the center and an area that encompasses much of the High Atlas Mountains and is characterized by high and medium altitudes. This position appears to offer a fertile ground for development of many leishmaniasis vectors and the emergence of various clinical forms of the disease.

Foci of anthroponotic cutaneous leishmaniasis, caused by *Leishmania tropica*, have been identified or confirmed to be active in many provinces of Marrakesh-Tansift-Al Haouz region, like Essaouira (Pratlong et al., 1991), Chichaoua (Guernaoui et al., 2005b), and Al Haouz (Ramaoui et al., 2008; Boussaa et al., 2009). In addition, Moroccan Ministry of Health (2014) records each year some cases of *L. infantum* HVL in Marrakesh-Tansift-Al Haouz region.

In Morocco, the coexistence of *L. tropica* and *L. infantum* was demonstrated in the Sefrou province (northern Morocco) by Hmamouch et al. (2014). Anthroponotic foci of *L. tropica* CL are found around Fes and Taza in the eastern central parts of Morocco, not far from *L. infantum* HVL foci (Rhajaoui et al., 2004). Furthermore, several cases of canine leishmaniasis caused by *L. tropica* have been reported in regions where canine leishmaniasis is caused by *L. infantum* (Rhajaoui et al., 2007).

It is necessary to caution that cases of canine visceral leishmaniasis by *L. tropica* were reported in northern Morocco (Taounate and Al Hoceima provinces) (Guessouss-Idrissi et al., 1997a; Lemrani et al., 2002). Here, we investigate *L. infantum* HVL cases, reported and confirmed by Moroccan Ministry of Health services (Moroccan Ministry of Health, 2014).

L. infantum is transmitted by sand flies belong to the subgenus *Larroussius*, which is *P. perniciosus*, *P. longicuspis* and *P. ariasi* (Guessous-Idrissi et al., 1997b; Benabdennbi et al., 1999; Pesson et al., 2004). Studying the phlebotomine vectors bionomics is an interesting tool. It contributes to the planning for the prevention and control against leishmaniasis particularly in the areas where the risk is significant.

Our entomological data showed that the interval of altitudes between 850 and 1155 m was favorable to sand fly activity. In contrast, *Larroussius* species prefer the highest altitude (917–1637 m) with a low positive correlation ($r = 0.22$). Knowing that, other factors probably act like climate (temperature, precipitation, humidity, etc), vegetation type and geology.

P. perniciosus was present and abundant in all investigated stations, especially in cow pen biotopes but only as 'atypical' form (PNA), which appears to be the only form found in southern Morocco (Benabdennbi et al., 1999; Pesson et al., 2004; Guernaoui et al., 2005a; Boussaa et al., 2008). Although in these regions, *P. perniciosus* was sympatric with *P. longicuspis*, particularly in the mountainous zones and in the plain, *P. longicuspis* was the most abundant (Boussaa et al., 2008). In our sample collection, *P. longicuspis* was sympatric with *P. perniciosus*, particularly in the areas of Yabora, Zaouiate Bouhouta and Ourika and was present and abundant in 90% of the stations investigated in the present study. It was the most anthroponotic *Larroussius* species and collected especially in domestic habitat.

The epidemiological role of *P. longicuspis* and atypical form of *P. perniciosus* (PNA), if any, remains to be elucidated. In contrast, the vectorial role of *P. ariasi* has been described in Northern Morocco, where it was found infected by *L. infantum* (Hamdani, 1999).

P. ariasi was less abundant compared to other *Larroussius* species (collected in study area), except in Yabora (1155 m) where it was collected in both domestic and cow pen biotopes. Similarly, Guernaoui et al. (2006) collected *P. ariasi* in altitude ranging between 1000 and 1400 m. Since 2003, Moroccan Ministry of Health recorded some *L. infantum* HVL cases (Table 1). Al Haouz (1250 m) and Chichaoua (1000 m) provinces, which characterized by highest altitude, showed the maximum cases. As we collected *P. ariasi* in highest altitude, we hypothesize its implication in the transmission of *L. infantum* HVL cases observed in Marrakesh-Tansift-Al Haouz region.

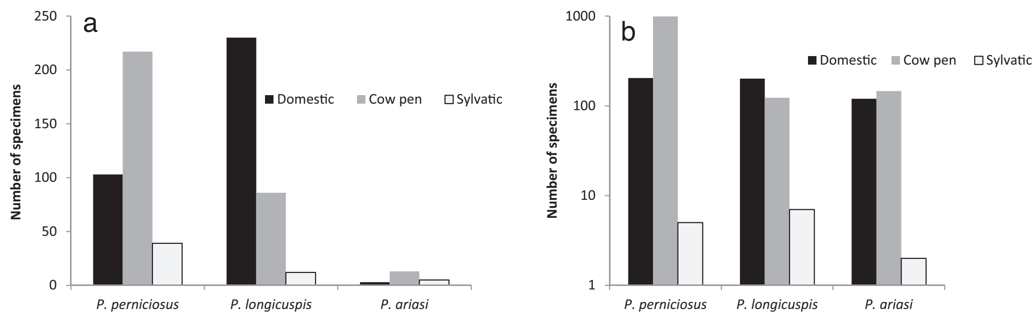

Fig. 2. Number of specimens of *Larroussius* species in various biotopes in (a) stations with altitude ≤900 and (b) stations with altitude >900.

At the same time, it is critically important to highlight the probable role as a vector for other species. In Iran, Rassi et al. (2013) found natural infection of *P. papatasi* (collected in 10/12 our stations), by *L. infantum*, with the absence of HVL cases, which means that this vector can transmit two species of *Leishmania* parasite (*L. major* and *L. infantum*) between human and reservoirs. *Sergentomyia* species may perhaps transmit the *Leishmania* parasite (Mukherjee et al., 1997; Parvizi and Amirkhani, 2008; Berdjane-Brouk et al., 2012). In this context, Senghor et al. (2011) suggests the possibility that certain species of *Sergentomyia* genus (41.46% of our caught) may play the role of vector of *L. infantum. P. alexandri* (collected in 5/12 our stations) could be also a HVL vector like in Sahel region and in Iran (Le Pont et al., 1993; Azizi et al., 2006).

During the same year of seasonality study (2012), the densities of *P. longicuspis, P. perniciosus* and *P. ariasi* showed an overlap between them (Fig. 3). In these three stations, *P. perniciosus* shows the high density while *P. ariasi* is characterized by the lower. Aransay et al. (2004), affirmed that *P. perniciosus* is widespread and less affected by climate conditions, while *P. ariasi* showed a preference for humid, cold areas. In Morocco, other researchers have also reported that *P. ariasi* generally thrives in humid, cold high-mountain areas (Rispail et al., 2002; Guernaoui et al., 2006).

The *Larroussius* activity varies notably over time (seasons) but also with space. In Ourika and Zaouiate Bouhouta stations (850 m and 969 m, respectively), *Larroussius* activity showed a bimodal seasonal trend with two abundance peaks, in summer (May–June–July) and in autumn (September–October). In Yabora (1155 m), *Larroussius* activity showed a mono-modal annual pattern with a long activity period extending from March to November with the highest density recorded in June (Fig. 3).

In Morocco, following the study of Faraj et al. (2013), *P. longicuspis* peaks in May and August at Ait Bousdouk (altitude of about 250 m; arid climate) and Azrou (1500 m; Saharan climate), respectively, while, *P. perniciosus* was more prevalent in September and October, respectively, at Bouassem (1100 m; subhumid climate) and Ait Chribou (1200 m; Semiarid climate). According to Guernaoui et al. (2005b), these *Larroussius* species showed a monophasic cycle in Chichaoua province (Lalla Aziza locality at 1148 m; semiarid climate), with one density peak in

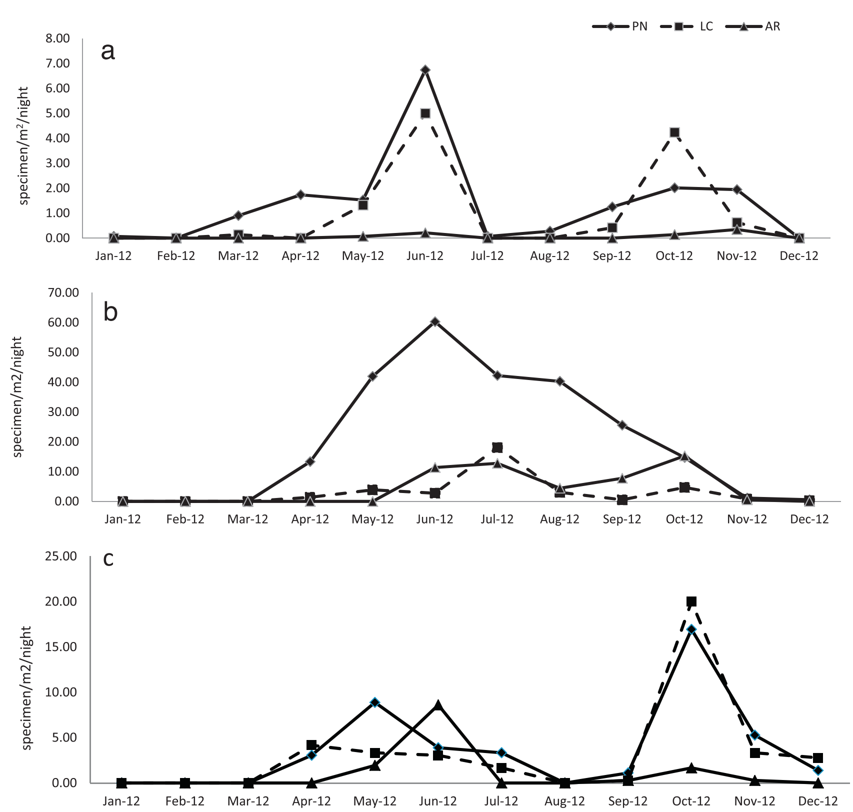

Fig. 3. Seasonal trends of *Phlebotomus perniciosus*, *P. longicuspis* and *P. ariasi* in (a) Ourika, (b) Yabora, and (c) Zaouiate Bouhouta recorded between January and December 2012.

August–September, September–October and October for, respectively, *P. longicuspis, P. perniciosus* and *P. ariasi.*

In Spain, many authors have noted a bimodal annual abundance pattern for *P. perniciosus* with two peaks observed (in July and in September) and a monophasic trend with one peak (in August) for *P. ariasi* (Morillas et al., 1982; Sanchis Marin et al., 1986; Lucientes Curdi et al., 1991; Gálvez et al., 2010).

5. Conclusion

Human visceral leishmaniasis is endemic in Morocco and constitutes a major public health problem. Perhaps, this form is sporadic but expanding to large parts of the country in all bioclimatic areas because of the abundance of its vectors: *P. perniciosus, P. ariasi* and *P. longicuspis*. So far, Marrakesh-Tensift-Al Haouz region has been epidemiologically underestimated.

This study constitutes a wake-up call and an update of the *L. infantum* HVL status in the region. Its geographical position, the presence of canine leishmaniasis infection (Dereure et al., 1986; Boussaa et al., 2014), the presence of many potential vectors with high density and a long activity period, the omnipresence of dogs–main *L. infantum* reservoir host–and the increase in regional temperatures can contribute to the spread and evolution of the incidence of HVL by *L. infantum*. Therefore, there is a need for a continuous surveillance in this potential *L. infantum* focus, especially in the period presenting a higher risk of disease transmission extending from March to November.

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References

Alvar, J., Vélez, I.D., Bern, C., Herrero, M., Desjeux, P., Cano, J., Jannin, J., den Boser, M., WHO Leishmaniasis Control Team, 2012. Leishmaniasis worldwide and global estimates of its incidence. PLoS ONE 7, e35671.

Amro, A., Hamdi, S., Lemrani, M., Idrissi, M., Hida, M., Rhajaoui, M., Hamarshe, O., Sconian, G., 2013. Moroccan *Leishmania infantum*: genetic diversity and population structure as revealed by multi-locus microsatellite typing. PLoS ONE 8, e77778.

Aransay, A.M., Testa, J.M., Morillas-Marquez, F., Lucientes, J., Ready, D.P., 2004. Distribution of sand fly species in relation to canine leishmaniasis from the Ebro Valley to Valencia, northeastern Spain. Parasitol. Res. 94, 416–420.

Azizi, K., Rassi, Y., Javadian, E., Motazedian, M.H., Rafizadeh, S., Yaghoobi Ershadi, M.R., Mohebali, M., 2006. Phlebotomus (*Paraphlebotomus*) *alexandri*: a probabl vector of *Leishmania infantum* in Iran. Ann. Trop. Med. Parasitol. 100, 63–68.

Benabdennbi, I., Pesson, B., 1998. A propos de la variabilité morphologique de *Phlebotomus perniciosus* (Diptera: Psychodidae). Bull. Soc. Fr. Parasitol. 16, 53–60.

Benabdennbi, I., Pesson, B., Cadi-soussi, M., Morillas, M.F., 1999. Morphological and isoenzymatic differentiation of sympatric populations of *P. perniciosus* and *P. longicuspis* (Diptera: Psychodidae) in northern Morocco. J. Med. Entomol. 36, 116–120.

Berdjane-Brouk, Z., Kone, A.K., Djimde, A.A., Charrel, R.N., Ravel, C., Delaunay, P., del Giudice, P., Diarra, A.Z., Doumbo, S., Goita, S., Thera, M.A., Depaquit, J., Marty, P., Doumbo, O.K., Izri, A., 2012. First detection of *Leishmania major* DNA in *Sergentomyia* (*Spelaeomyia*) darling from Cutaneous Leishmaniasis Foci in Mali. PLoS ONE 7, e28266.

Boussaa, S., Kasbari, M., El Mzabi, A., Boumezzough, A., 2014. Epidemiological investigation of canine leishmaniasis in Southern Morocco. Adv. Epidemiol. 2014, 8, http://dx.doi.org/10.1155/2014/104697 (ID 104697).

Boussaa, S., Pesson, B., Boumezzough, A., 2009. Faunistic study of the sandflies (Diptera: Psychodidae) in an emerging focus of cutaneous leishmaniasis in Al Haouz province, Morocco. Ann. Trop. Med. Parasitol. 103, 73–83.

Boussaa, S., Boumezzough, A., Remy, P.E., Glasser, N., Pesson, B., 2008. Morphological and isoenzymatic differentiation of *Phlebotomus perniciosus* and *Phlebotomus longicuspis* (Diptera: *Psychodidae*) in Southern Morocco. Acta Trop. 106, 184–189.

Dereure, J., Velez, I.D., Pratlong, F., Denial, M., Lardi, M., Moreno, G., Serres, E., Lanotte, G., Rioux, J.A., 1986. La leishmaniose viscérale autochtone au Maroc méridional. Présence de Leishmania infantum MON-1 chez le Chien en zone présaharienne. In Leishmania. Taxonomie et Phylogenèse. Applications Eco-épidémiologiques, Int Coll CNRS/INSERM/OMS (2–6 July 1984). Institut Méditerranéen d'Etudes Epidémiologiques et Ecologiques, Montpellier, pp. 421–425.

Faraj, C., El Bachir, A., El Rhazi, M., Ouahabi, S., Lakraa, L., El Kohli, M., Ameur, B., 2013. Distribution and bionomic of sand flies in five ecologically different cutaneous leishmaniasis foci in Morocco. ISRN Epidemiol. 2013, 8, http://dx.doi.org/10.5402/2013/145031 (Article ID 145031).

Gálvez, R., Descalzo, M.A., Miró, G., Jiménez, M.I., Martín, O., Dos Santos-Brandao, F., Guerrero, I., Cubero, E., Molina, R., 2010. Seasonal trends and spatial relations between environmental/meteorological factors and leishmaniosis sand fly vector abundances in Central Spain. Acta Trop. 115, 95–102.

Guernaoui, S., Boumezzough, A., Laamrani, A., 2006. Altitudinal structuring of sand flies (Diptera: Psychodidae) in the High-Atlas mountains (Morocco) and its relation to the risk of leishmaniasis transmission. Acta Trop. 97, 346–351.

Guernaoui, S., Pesson, B., Boumezzough, A., Pichon, G., 2005a. Distribution of phlebotomine sandflies, of the subgenus *Larroussius*, in Morocco. Med. Vet. Entomol. 19, 111–115.

Guernaoui, S., Boumezzough, A., Pesson, B., Pichon, G., 2005b. Entomological investigations in Chichaoua: an emerging epidemic focus of cutaneous leishmaniasis in Morocco. J. Med. Entomol. 42, 697–701.

Guessouss-Idrissi, N., Berrag, B., Riyad, M., Sahibi, H., Bichichi, M., Rhalem, A., 1997a. *Leishmania tropica*: etiologic agent of a case of visceralizing canine leishmaniasis in north Morocco. Am. J. Trop. Med. Hyg. 57, 172–173.

Guessous-Idrissi, N., Hamdani, A., Rhalem, A., Riyad, M., Sahibi, H., Dehbi, F., Bichichi, M., Essari, A., Berrag, B., 1997b. Epidemiology of human visceral leishmaniasis inTaounate, a northern province of Morocco. Parasite 2, 181–185.

Hamdani, A., 1999. Etude de la faune phlébotomienne dans trois foyers de leishmanioses au Nord du Maroc: Espèces, Abondance, Saisonalité et incrimination du vecteur. Faculté des Sciences Semlalia, Marrakech, pp. 57 (Thèse de 3ème cycle).

High Planning Commission of Morocco (HCP), 2014. Report of General Census of Population and Housing of 2004. High Planning Commission of Morocco (HCP), Morocco, ⟨http://www.hcp.ma/Recensement-General-de-la-Population-et-de-l-Habitat-2004_a92.html⟩.

Hmamouch, A., Amarir, F., Fellah, H., Karzaz, M., Bekhti, K., Rhajaoui, M., Sebti, F., 2014. Coexistence of *Leishmania tropica* and *Leishmania infantum* in Sefrou province, Morocco. Acta Trop. 130, 94–99.

Kahime, K., Boussaa, S., Bounoua, L., Fouad, O., Messouli, M., Boumezzough, A., 2014. Leishmaniasis in Morocco: diseases & vectors. Asian Pac. J. Trop. Dis. 4, 530–534.

Killick Kendrick, R., Tang, Y., Killick Kendrick, M., Sang, D.K., Sirdar, M.K., Ke, L., Ashford, R.W., Schorscher, J., Johnson, R.H., 1991. The identification of female sandflies of the subgenus *Larroussius* by the morphology of the spermathecal ducts. Parasitologia 33, 335–347.

Lemrani, M., Nejjar, M., Pratlong, F., 2002. A new *Leishmania tropica* zymodeme–causitive agent of canine visceral leishmaniasis in northern Morocco. Ann. Trop. Med. Parasitol. 96, 637–638.

Léger, N., Pesson, B., Madulo-Leblond, G., Abonnenc, E., 1983. Differentiation of females of the subgenus *Larroussius* Nitzulescu 1931 (Diptera-Phlebotomidae) of the Mediterranean region. Ann. Parasitol. Hum. Comp. 58, 611–623.

Le Pont, F., Robert, V., Vattier-Bernard, G., Rispail, P., Jarry, D., 1993. Notes sur les phlebotomes de l'Aïr (Niger). Bull. Soc. Pathol. Exot. 86, 286–289.

Lucientes Curdi, J., Benito de Martin, M.I., Castillo Hernandez, J.A., Orcajo Teresa, J., 1991. Seasonal dynamics of *Larroussius* species in Aragon (N.E.Spain). Parassitologia 33, 381–386.

Morillas, M.F., Benitez, D.C.G., Collando, J.G., Ontiveros, J.M.U., 1982. Presencia en España de *Phlebotomus* (*Larroussius*) longicuspis Nitzulescu, 1930. Rev. Eberica Parasitol. Extra, 191–196.

Moroccan Ministry of Health, 2014. The HVL due to *L. infantum* Data is Provided by the Directorate of Epidemiology and Disease Control. Moroccan Ministry of Health, Rabat, Morocco.

Moroccan Ministry of Health, 2011. A Report on Progress of Control Programs Against Parasitic Diseases. Directorate of Epidemiology and Disease Control, Ministry of Health, Rabat, Morocco (2013 July 19). Available from: ⟨http://www.sante.gov.ma/departements/delm/index-delm.htm⟩.

Moroccan Ministry of Health, 2010. Lutte contre les leishmanioses: Guide des activités. Moroccan Ministry of Health, pp. 120.

Moroccan Ministry of Health, 1997. A Report on Progress of Control Programs Against Parasitic Diseases. Directorate of Epidemiology and Disease Control, Ministry of Health, Rabat, Morocco, pp. 53–57, ⟨http://www.sante.gov.ma/departements/delm/index-delm.htm⟩.

Mukherjee, S., Quamarul, M.H., Ghosh, A., Kashi, N.G., Bhattacharya, A., Adhya, S., 1997. *Leishmania* DNA in *Phlebotomus* and *Sergentomyia* species during a Kala-azar epidemic. Am. J. Trop. Med. Hyg. 4, 423–425.

Parvizi, P., Amirkhani, A., 2008. Mitochondrial DNA characterization of *Sergentomyia sintoni* populations and finding mammalian Leishmania infectionsin this sandfly by using ITS-rDNA gene. Iran. J. Vet. Res. 22, 9–18.

Pesson, B., Ready, J.S., Benabdennbi, I., Martin-Sanchez, J., Esseghir, S., Cadi Soussi, M., Morillas-Marquez, F., Ready, P.D., 2004. Sandflies of the *Phlebotomus perniciosus* complex:mitochondrial introgression and a new sibling species of *P. longicuspis* in the Moroccan Rif. Med. Vet. Entomol. 18, 25–37.

Pratlong, F., Rioux, J.A., Dereure, J., Mahjour, J., Gallego, M., Guilvard, E., Lanotte, G., Périères, J., Martini, A., Saddiki, A., 1991. *Leishmania tropica* au Maroc. IV. Diversité isozymique intrafocale. Ann. Parasitol. Hum. Comp. 66, 100–104.

Ramaoui, K., Guernaoui, S., Boumezzough, A., 2008. Entomological and epidemiological study of a new focus of cutaneous leishmaniasis in Morocco. Parasitol. Res. 103, 859–863.

Rassi, Y., Karami, H., Abai-Mohammad, R., Mohebali, M., Bakshi, H., Oshaghi, M.A., Rafizadeh, S., Bagherpoor, H.H., Hosseini, A., Gholami, M., 2013. First detection of

Leishmania infantum DNA in wild caught *Phlebotomus papatasi* in endemic focus of cutaneous leishmaniasis, South of Iran. Asian Pac. J. Trop. Biomed. 3, 825–829.
Ready, P.D., 2014. Epidemiology of visceral leishmaniasis. Clin. Epidemiol. 6, 147–154.
Rhajaoui, M., 2009. Human leishmaniases in Morocco: a nosogeographical diversity. Pathol. Biol. 59, 226–229.
Rhajaoui, M., Abedelmajeed, N., Fellah, H., Azmi, K., Amrir, F., Al-jawabreh, A., Ereqat, S., Planer, J., Abdeen, Z., 2007. Molecular typing reveals emergence of a new clinic-epidemiologic profile of cutaneous leishmanisis in Morocco. Emerg. Infect. Dis. 13, 1358–1360.
Rhajaoui, M., Fellahn, H., Pratlong, F., Dedet, J.P., Lyagoubi, M., 2004. Leishmaniasis due to *Leishmania tropica* MON-102 in a new Moroccan focus. Trans. R. Soc. Trop. Med. Hyg. 98, 299–301.
Rioux, J.A., Mahjour, J., Gallego, M., Dereure, J., Périères, J., Laamrani, A., Riera, C., Saddiki, A., Mouki, B., 1996. Leishmaniose cutanée humaine à *Leishmania infantum* MON-24 au Maroc. Bull. Soc. Fr. Parasitol. 14, 179–183.
Rispail, P., Dereure, J., Jarry, D., 2002. Risk zones of human leishmaniases in the western Mediterranean basin. Correlations between vectors sand flies, bioclimatology and phytosociology. Oswaldo Cruz, Rio de Janeiro 97, 477–483.
Sanchis Marin, M.C., Morillas-Marquez, F., Gonzalez-Castro, J., Benavides-Delgado, I., Reyes Magana, A., 1986. Dinamica estacional de los flebotomos (Diptera: Psychodidae) de la provincia de Almeria (Espana). Rev. Iber. Parasitol. 46, 285–291.
Senghor, M.W., Faye, M.N., Faye, B., Diarra, K., Elguero, E., Gaye, O., Bañuls, A.L., Niang, A.A., 2011. Ecology of Phlebotomine Sand Flies in the Rural Community of Mont Rolland (Thies Region, Senegal): area of Transmission of Canine Leishmaniasis. PLoS ONE 6, e14773.
Tamimy, H., 2011. The Infantile Visceral Leishmaniasis (About 73 Cases). Sidi Mohammed Ben Abdellah University, Faculty of Medicine and Pharmacy (Ph.D. in Medicine). ⟨http://scolarite.fmp-usmba.ac.ma/cdim/mediatheque/e_theses/89-11.pdf⟩.